

Approximation Algorithms for Matroid-Intersection Coloring with Applications to Rota's Basis Conjecture

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Matroid Basics

Matroid Definition

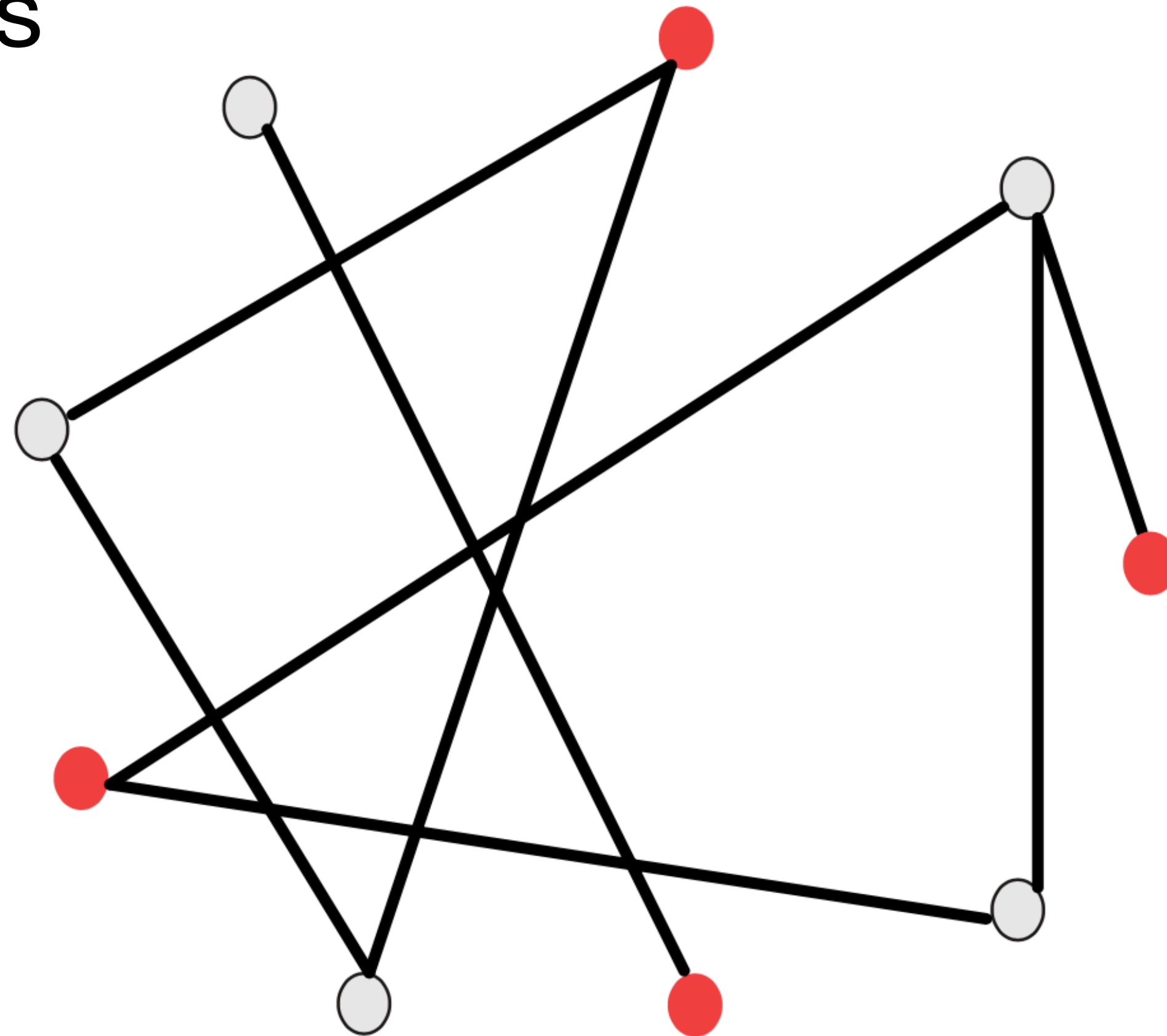
- **Def:** A matroid is $M = (E, \mathcal{S})$ where...
 - E is the set of elements, called the **ground set**.
 - $\mathcal{S}$ is a family of subsets of E , called the **independent sets**.
 - **Subset Property:** If $I \in \mathcal{S}$ and $J \subseteq I$, then $J \in \mathcal{S}$.
 - **Exchange Property:** If $I, J \in \mathcal{S}$ and I is larger than J , then there is some $x \in I \setminus J$ such that $J \cup \{x\} \in \mathcal{S}$.

Importance of Matroids

- **Fundamental objects** in combinatorial optimization
- **Simplify and generalize** combinatorial results
- Extensively studied in the “ACO” literature

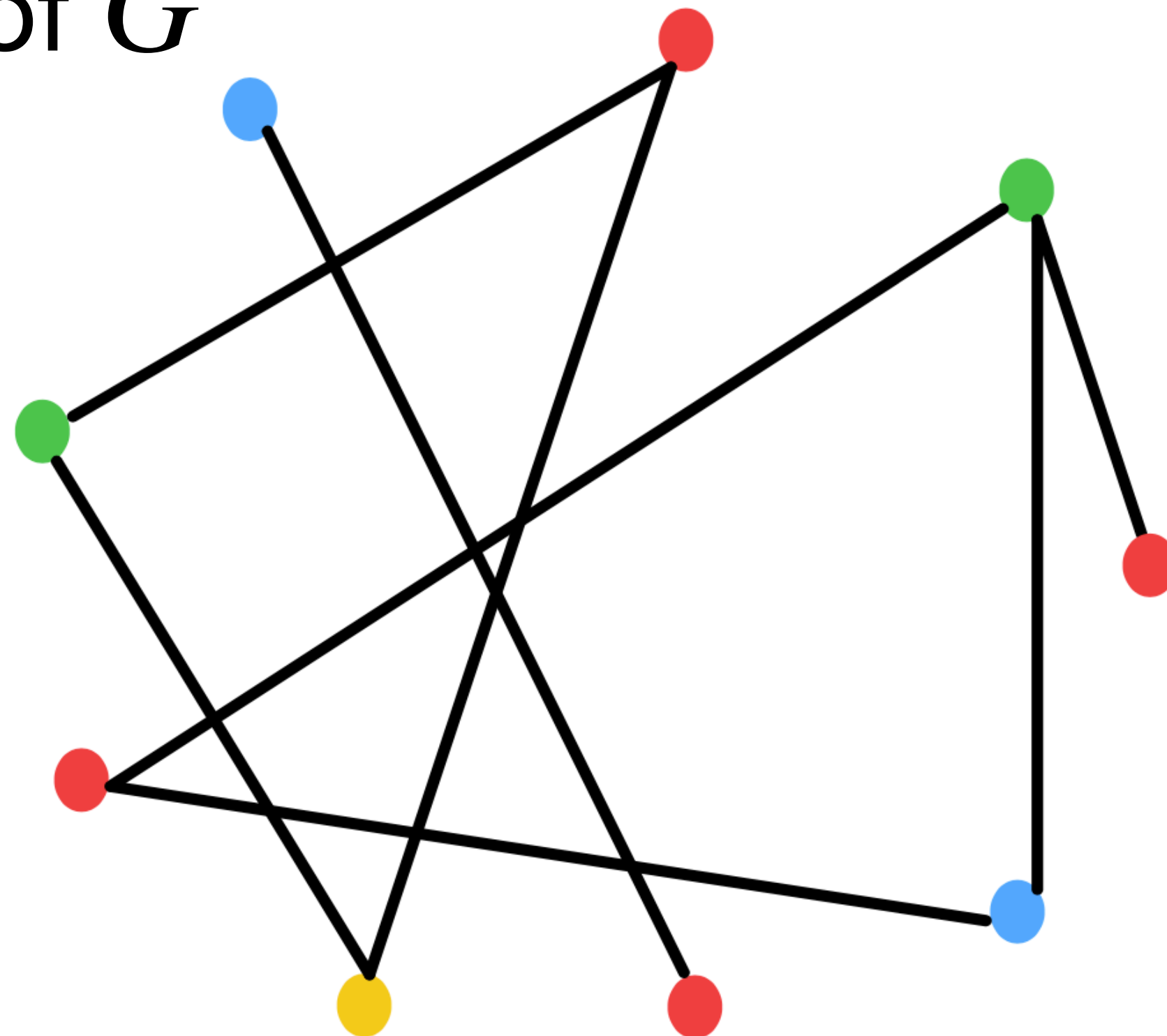
Chromatic Number

- **Given:** A set system $N = (E, \mathcal{S})$ with the **subset property**.
 - **Ex.** Stable sets in a graph G .



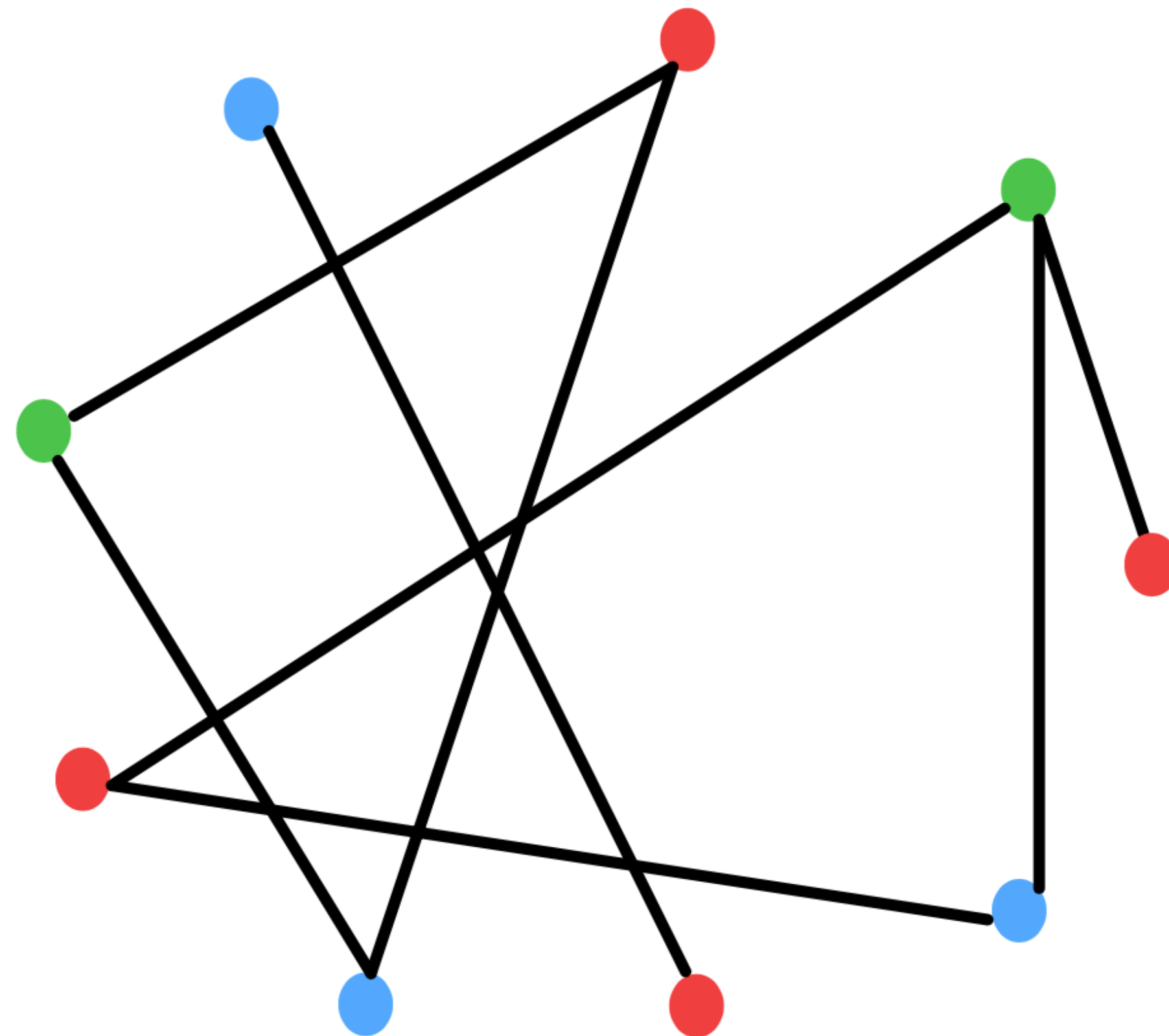
Chromatic Number

- **Def:** A **valid coloring** of $N = (E, \mathcal{S})$ is a coloring of the elements such that each **color class** is in $\mathcal{S}$.
 - **Ex.** 4-coloring of G



Chromatic Number

- **Def:** The **chromatic number** $\chi(N)$ of $N = (E, \mathcal{S})$ is the **minimum** number of colors in any valid coloring.
 - **Ex.** $\chi(G) = 3$



Matroid-Intersection Coloring Problem

Matroid-Intersection Coloring Problem

The Problem

- **Given:** Two matroids $M_1 = (E, \mathcal{S}_1)$ and $M_2 = (E, \mathcal{S}_2)$
- **Goal:** Compute a minimum valid coloring of $M_1 \cap M_2$
- **OPT:** $\chi(M_1 \cap M_2)$

Key Results

- **One Matroid:** Polynomial-time solvable [**Edm68**, **Sch03**]
- **Two Matroids:** NP-Hard [**BS21**, **BCK23**, **HKK24**]
- Approximation algorithms

Our Contribution

1. **2-approximation** for Matroid-Intersection Coloring Problem
 - Deterministic algorithm
 - **General Result:** $(k^2 - k)$ -approximation for k -MICP
 - No $O(1)$ -approximation was known even for two matroids

Our Contribution

1. **2-approximation** for Matroid-Intersection Coloring Problem
2. **$(1 + o(1))$ -approximation** for MICP when χ is large
 - Randomized algorithm
 - Two Matroids
 - Poly-time algorithm for Asymptotic **Rota's Basis Conjecture**

Our Contribution

1. **2-approximation** for Matroid-Intersection Coloring Problem
 2. **$(1 + o(1))$ -approximation** for MICP when χ is large
- **Main Insight: Polyhedral view** of problem

2-Approximation for MICP

Understanding the Literature

- Types of Matroids
 - **Combinatorial:** Partition, Graphic, Transversal, Laminar...
 - **General:** All matroids
- Types of Proofs
 - **Nonconstructive:** Does **not** yield a poly-time algorithm.
 - **Constructive:** Does yield a poly-time algorithm.

Understanding the Literature

Matroids

General

Combinatorial

Nonconstructive

Constructive

Proofs

Understanding the Literature

Matroids

General

Combinatorial

Nonconstructive

Constructive

Proofs

Understanding the Literature

Matroids

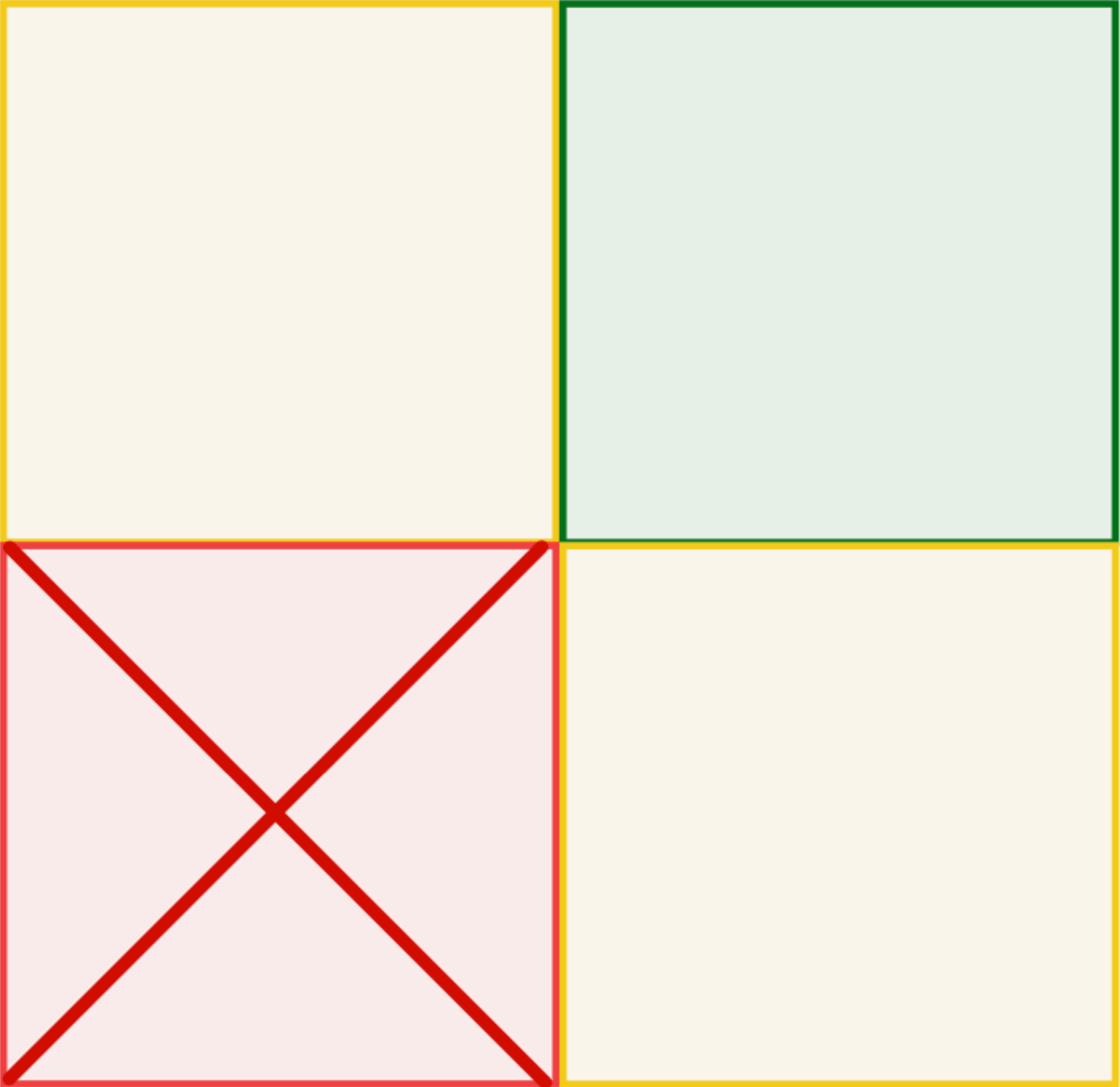
General

Combinatorial

Nonconstructive

Constructive

Proofs

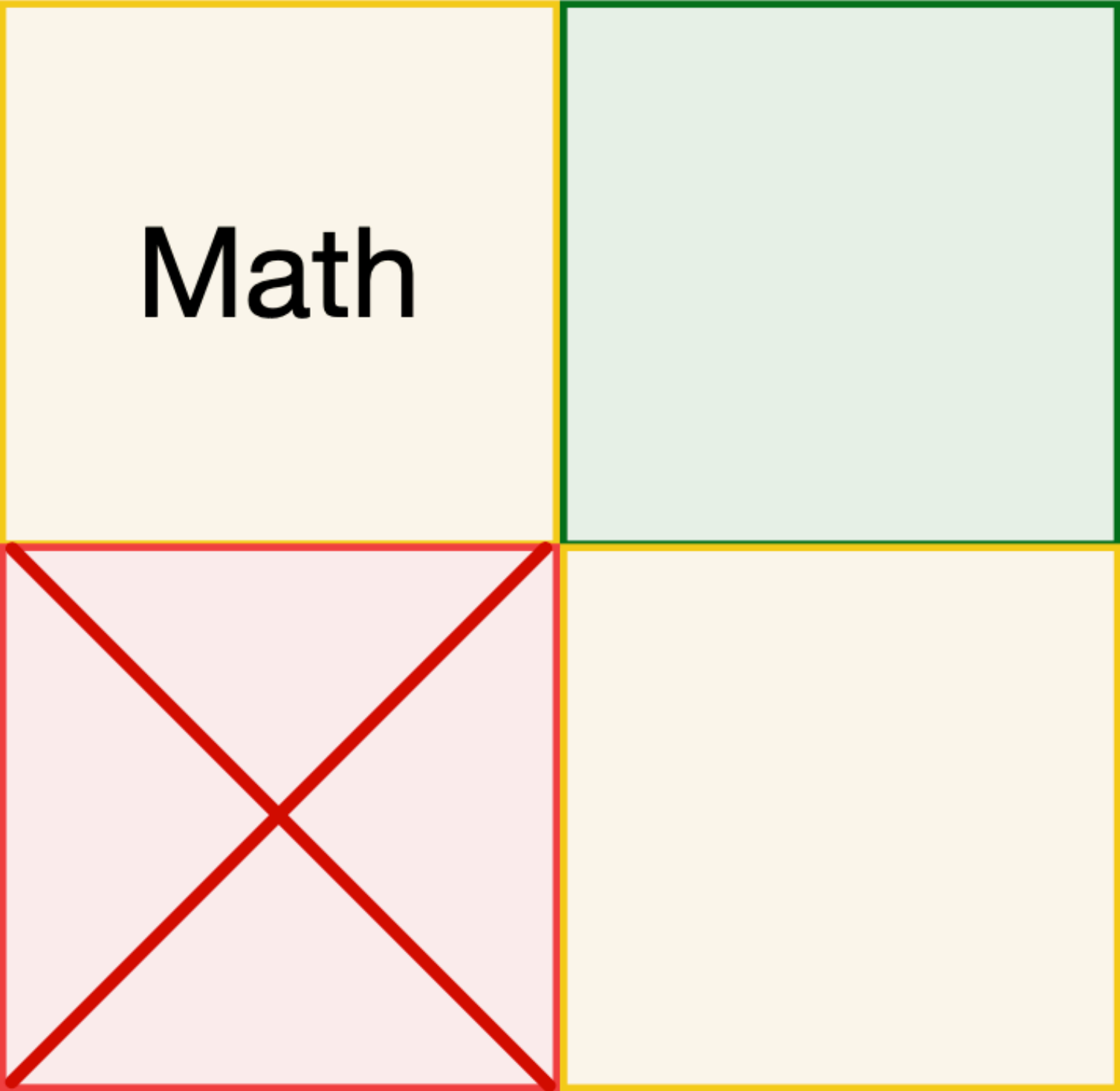


Understanding the Literature

Matroids

General

Combinatorial



Nonconstructive

Constructive

Proofs

Math: **General** and **Nonconstructive**

- 10-100 papers (many **combinatorial** and **nonconstructive**)
- **Thm ([AB06]):** For two **general** matroids M_1 and M_2 ,

$$\chi(M_1 \cap M_2) \leq 2 \max\{\chi(M_1), \chi(M_2)\}$$

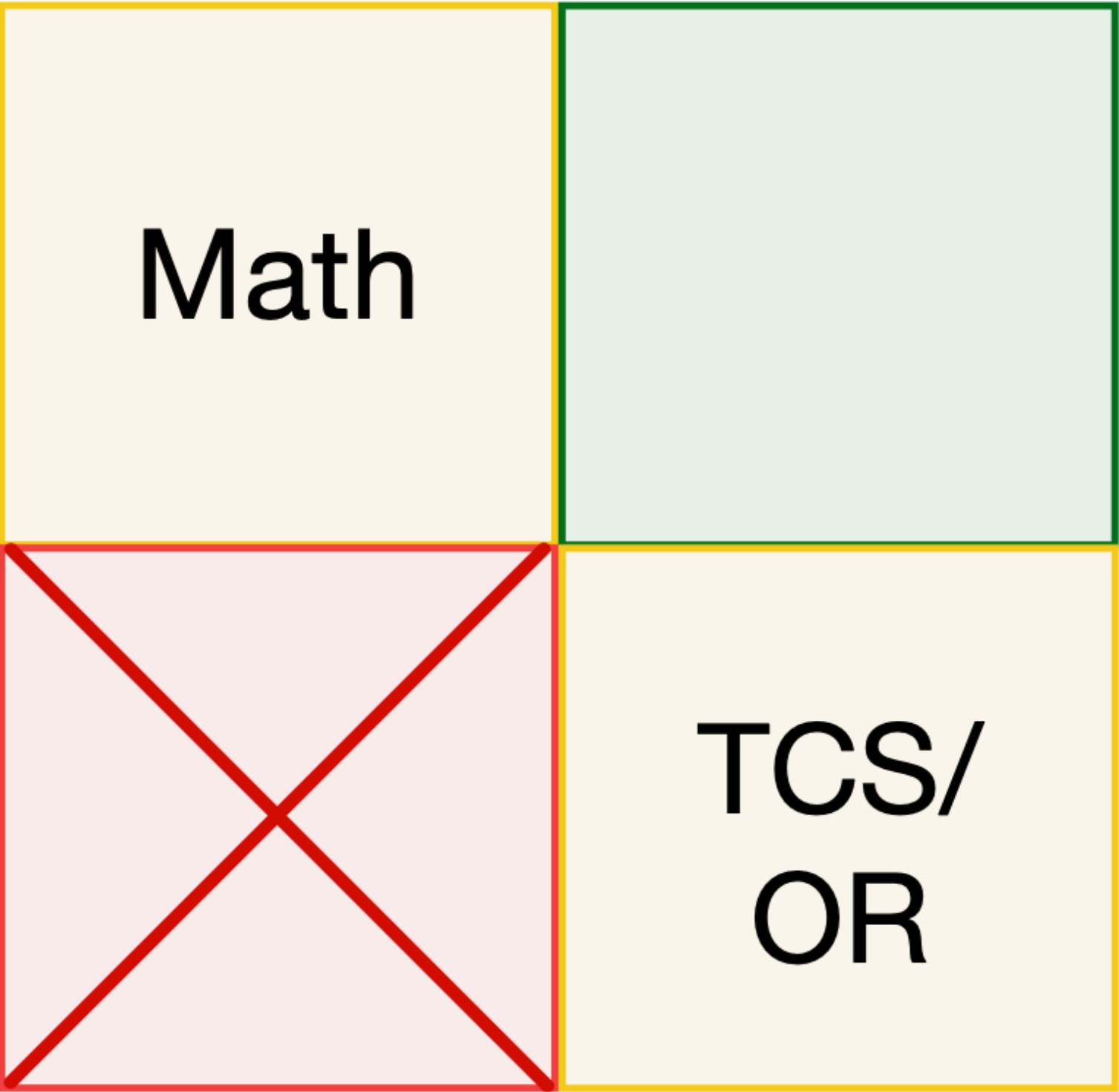
- Topological methods and Sperner's Lemma
- **Nonconstructive**

Understanding the Literature

Matroids

General

Combinatorial



Nonconstructive

Constructive

Proofs

TCS/OR: Combinatorial and Constructive

- 5-10 papers
- **One General Matroid ([Edm68, Sch03])**
 - Polynomial-time algorithm to **exactly** compute $\chi(M)$.
 - Find augmenting paths in an exchange graph G .
- **Two Partition Matroids ([Kön16])**
 - Polynomial-time algorithm to **exactly** compute $\chi(M_1 \cap M_2)$.
 - Reduce to edge coloring a bipartite graph.

TCS/OR: **Combinatorial** and **Constructive**

- **Two Combinatorial Matroids** ([IMP21, BSY21, LMP21])
 - Partition reduction technique
 - **Types:** Graphic, Transversal, Laminar, Gammoid...
 - Convert matroid into partition matroid with factor-of-2 blowup in the chromatic number.
 - **Conj** ([BSY21]): Every matroid is reducible in this way.
 - **Thm** ([LMP22, AKKG23]): Binary matroids are not.

TCS/OR: **Combinatorial** and **Constructive**

- **One General + One Combinatorial Matroid ([AMPZ26])**
 - Augmenting paths technique
 - Extension of Edmonds' Algorithm for One General Matroid ([Edm68, Sch03])
 - Cannot handle two general matroids



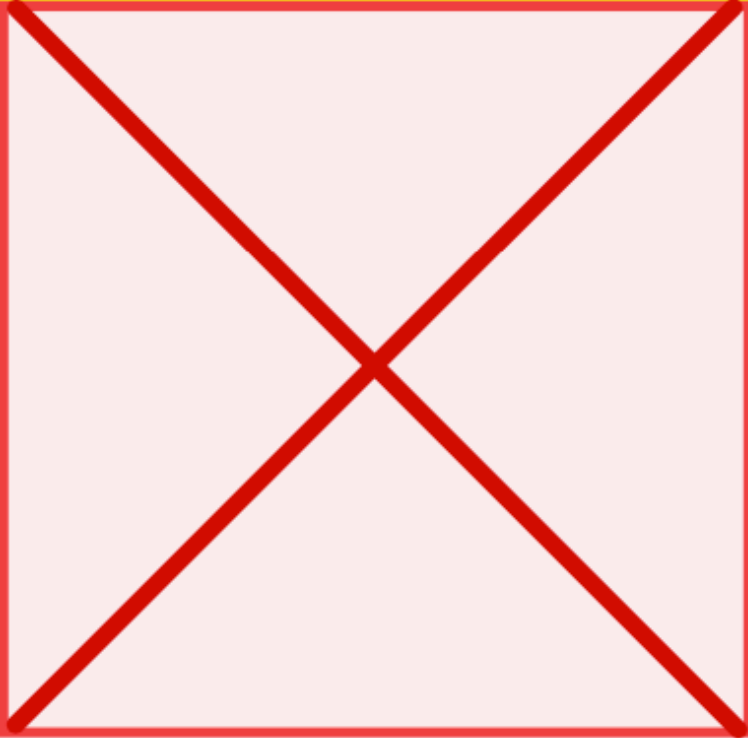
To appear
IPCO 2026!

Understanding the Literature

Matroids

General

Combinatorial

Math	?
	TCS/ OR

Nonconstructive

Constructive

Proofs

?: General and Constructive

- 0 papers
- **Two General Matroids**
 - $O(\log n)$ -approximation algorithm via **Set Cover**
 - Bad approximation factor 😞
 - Does not connect $\chi(M_1 \cap M_2)$ to $\chi(M_1)$ and $\chi(M_2)$ 😞

?: General and Constructive

- **Thm ([AB06]):** For two **general** matroids M_1 and M_2 ,

$$\chi(M_1 \cap M_2) \leq 2 \max\{\chi(M_1), \chi(M_2)\}$$

- **Research Question 1:** Does there exist a **constructive** proof (polynomial-time algorithm) for

$$\chi(M_1 \cap M_2) \leq 1000 \max\{\chi(M_1), \chi(M_2)\}?$$

$$\chi(M_1 \cap M_2) \leq 2 \max\{\chi(M_1), \chi(M_2)\}?$$

Our Contribution

Our Contribution

- **Thm:** For two **general** matroids M_1 and M_2 , there is a **polynomial-time** algorithm that computes a
$$2 \max \{ \chi(M_1), \chi(M_2) \} \text{-coloring of } M_1 \cap M_2.$$
- First polynomial-time $O(1)$ -approximation algorithm for the Matroid-Intersection Coloring Problem
- **Constructive** proof of $\chi(M_1 \cap M_2) \leq 2 \max \{ \chi(M_1), \chi(M_2) \}$

Technical Overview

- **Polyhedral view** of problem
 1. **Standard:** Reduce to 3-matroid intersection problem.
 2. **Iterative Rounding:** Produces a “pseudocoloring”.
 3. **Graph Coloring:** Build conflict graph and compute a valid coloring.
- We will focus on **Step 3**.

Overview of Step 3: Graph Coloring

- Definitions
 - Flexible decomposition
 - Pseudocoloring
- Pseudocoloring Lemma
 - Output of Step 2 (Iterative Rounding)
- Proof of Theorem via Graph Coloring

Flexible Decomposition

Circuit:

Minimal dependent set

- **Relaxation** of independence
- **Def:** For a general matroid $M = (E, \mathcal{S})$ and subset $S \subseteq E$, a **flexible decomposition** is a **partition** $P_1, P_2, \dots, P_k$ of S such that...
 1. Each part P_i is an **independent set** or a **circuit**.
 2. For any collection of **independent sets** $I_i \subseteq P_i$ for $i \in [k]$, their **union** $\bigcup I_i$ is **independent**.

Pseudocoloring

- **Relaxation** of valid coloring
- **Def:** For two general matroids M_1 and M_2 ,
 - A **valid coloring** of $M_1 \cap M_2$ is a coloring where the color classes are **independent sets** in M_1 and M_2 .
 - A **pseudocoloring** of $M_1 \cap M_2$ is a coloring where the color classes have a **flexible decomposition** in M_1 and M_2 .

Pseudocoloring Lemma

- **Lemma:** For two general matroids M_1 and M_2 , there is a polynomial-time algorithm that computes a

$\max\{\chi(M_1), \chi(M_2)\}$ -pseudocoloring of $M_1 \cap M_2$.

- **Thm:** For two general matroids M_1 and M_2 , there is a polynomial-time algorithm that computes a

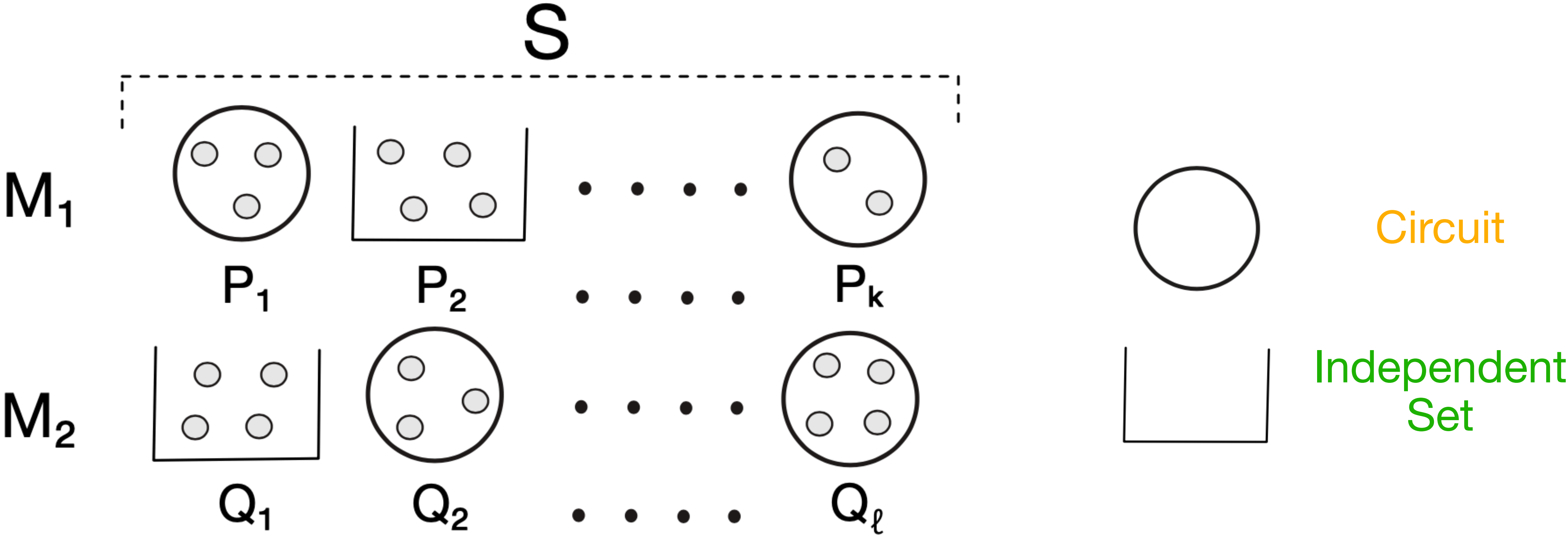
$2 \max\{\chi(M_1), \chi(M_2)\}$ -valid-coloring of $M_1 \cap M_2$.

Proof of Main Theorem

- **Goal:** Replace each pseudo color class with 2 valid color classes.
- Consider a pseudo color class S
- S has a flexible decomposition $P_1, P_2, \dots, P_k$ in M_1 and $Q_1, Q_2, \dots, Q_\ell$ in M_2 .

Proof of Main Theorem

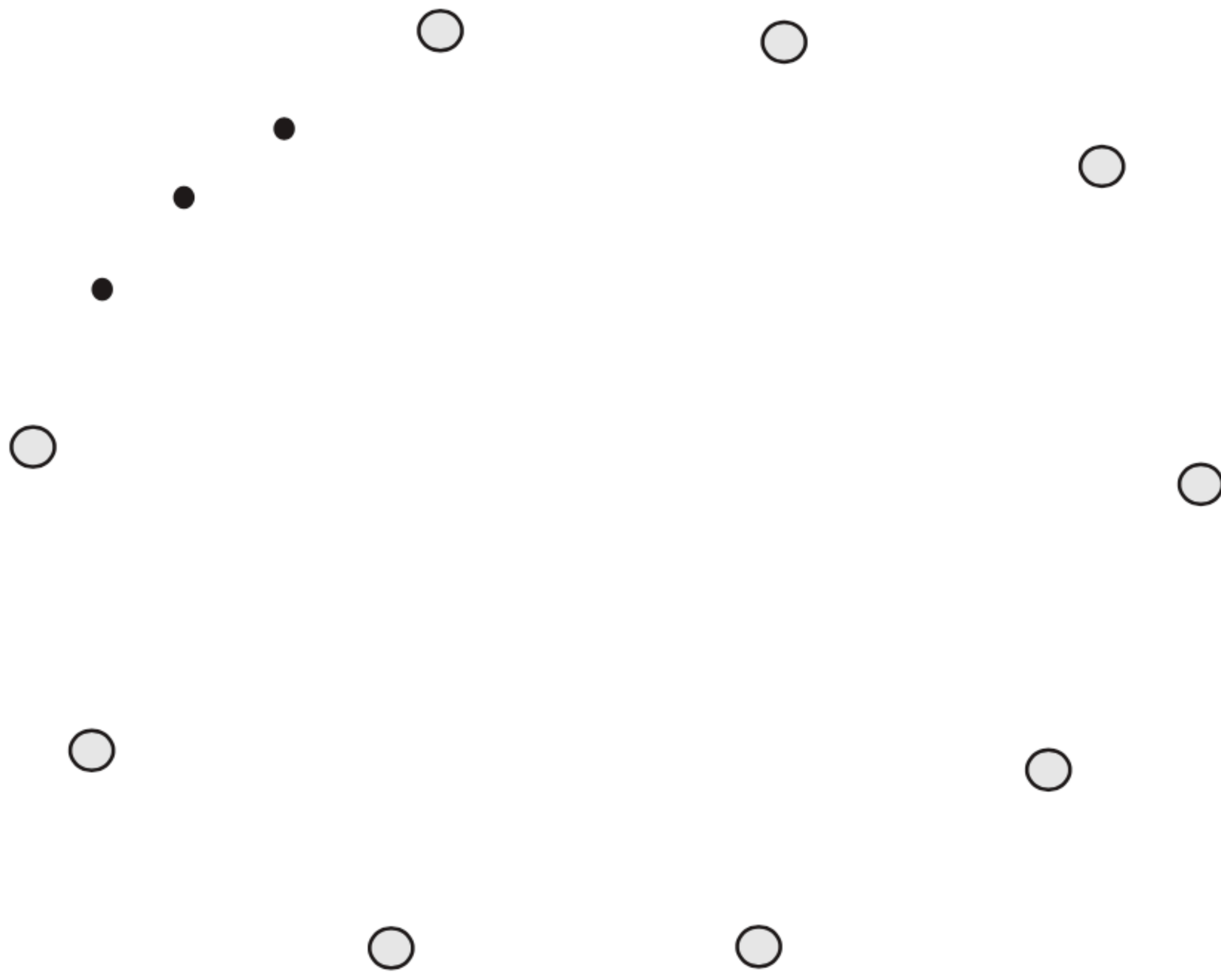
- **Goal:** Replace each **pseudo color class** with 2 **valid color classes**.

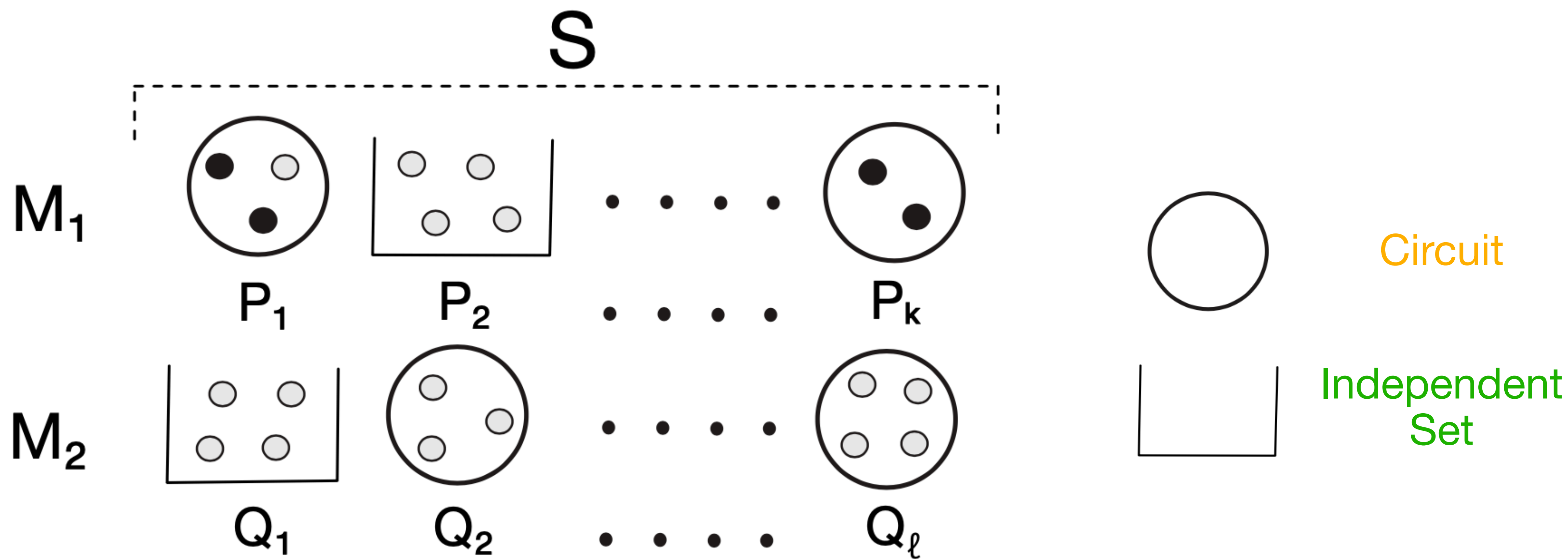


Proof of Main Theorem

- Build a conflict graph G
 - **Vertices:** S
 - **Edges:** For each **circuit** part P_i , add an edge between two arbitrary elements in P_i . Do the same for each **circuit** part Q_j .

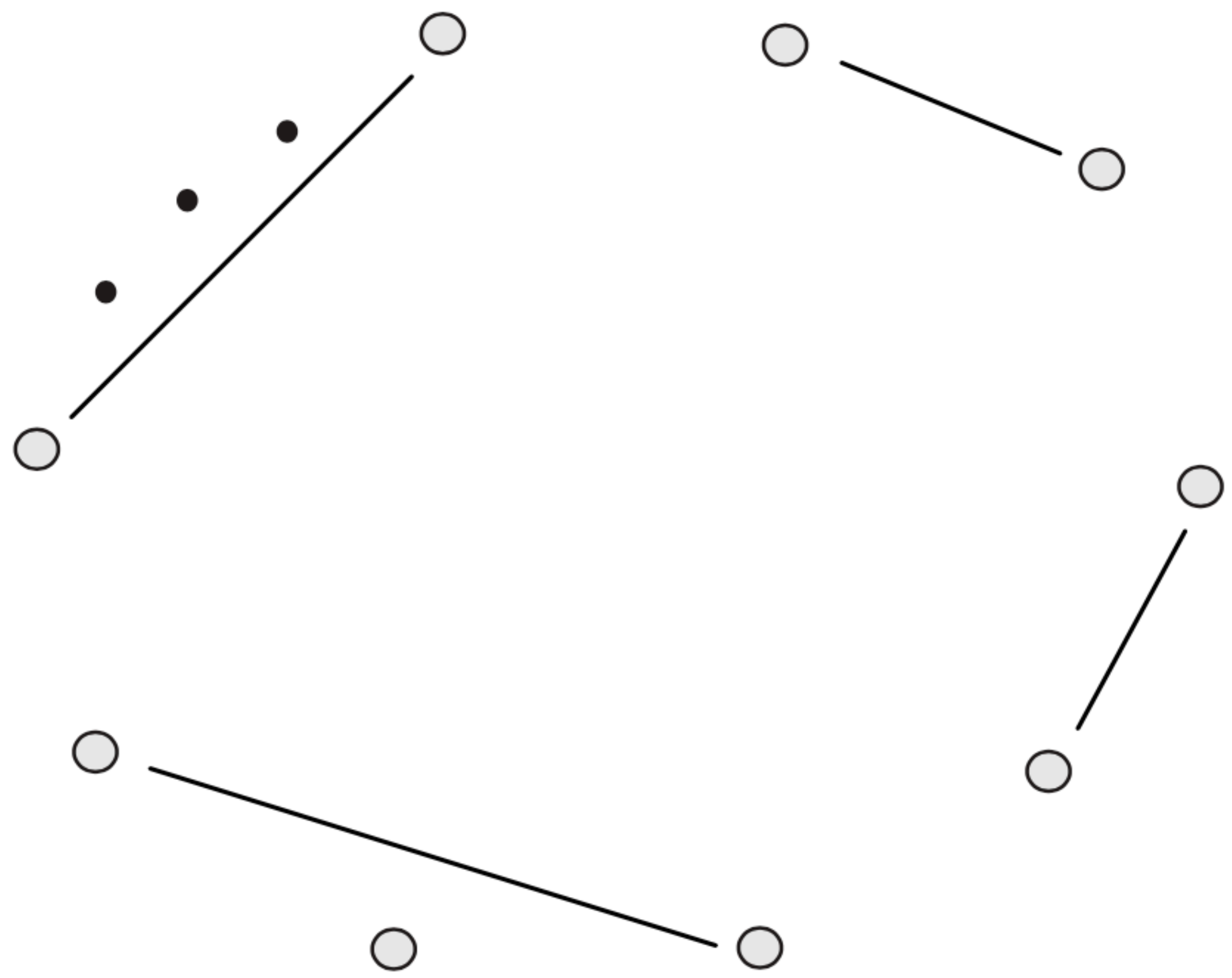
G

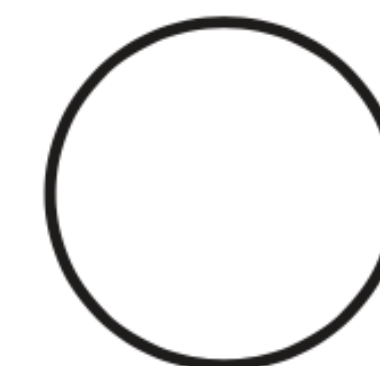
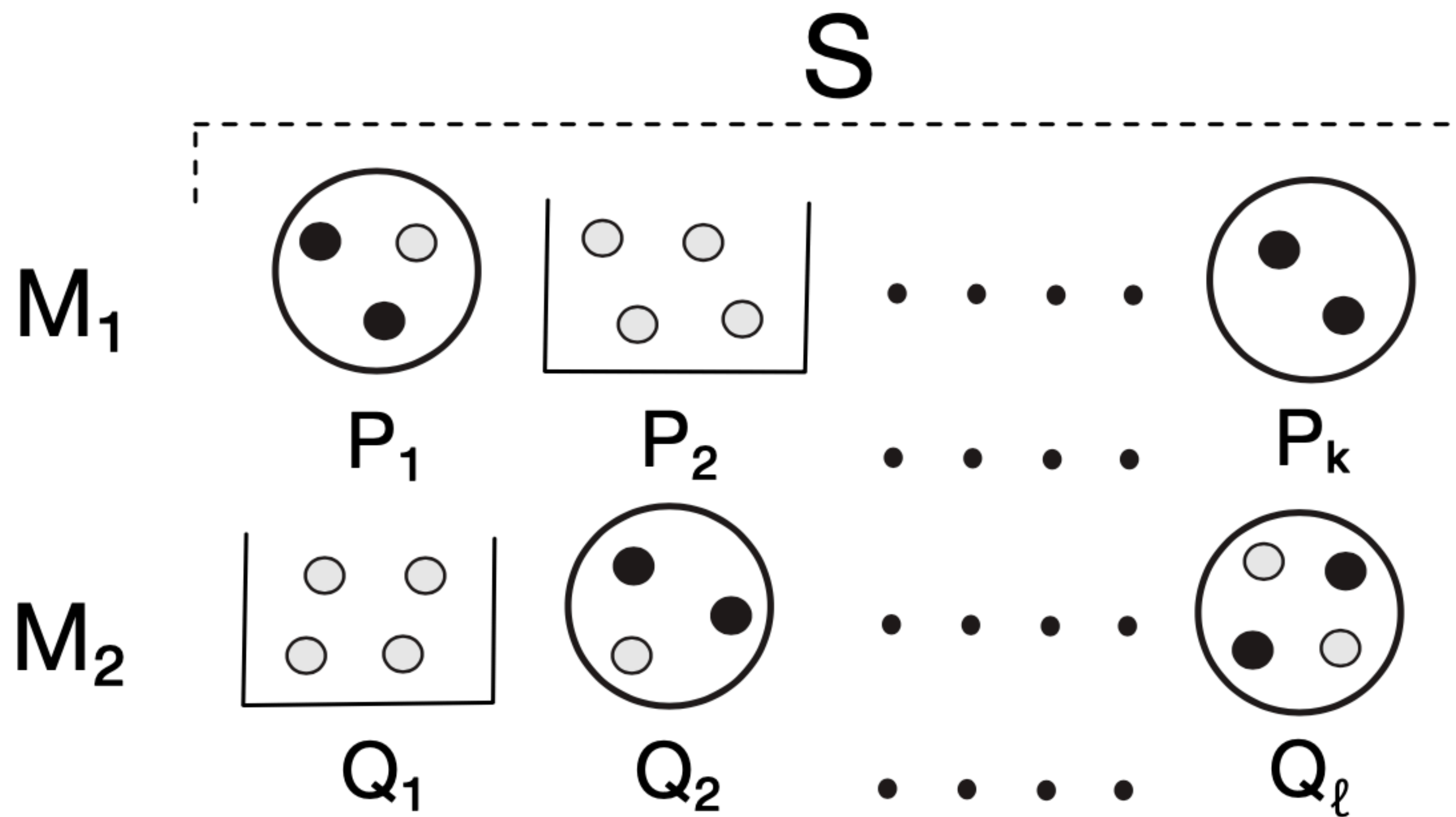




G

M_1 —



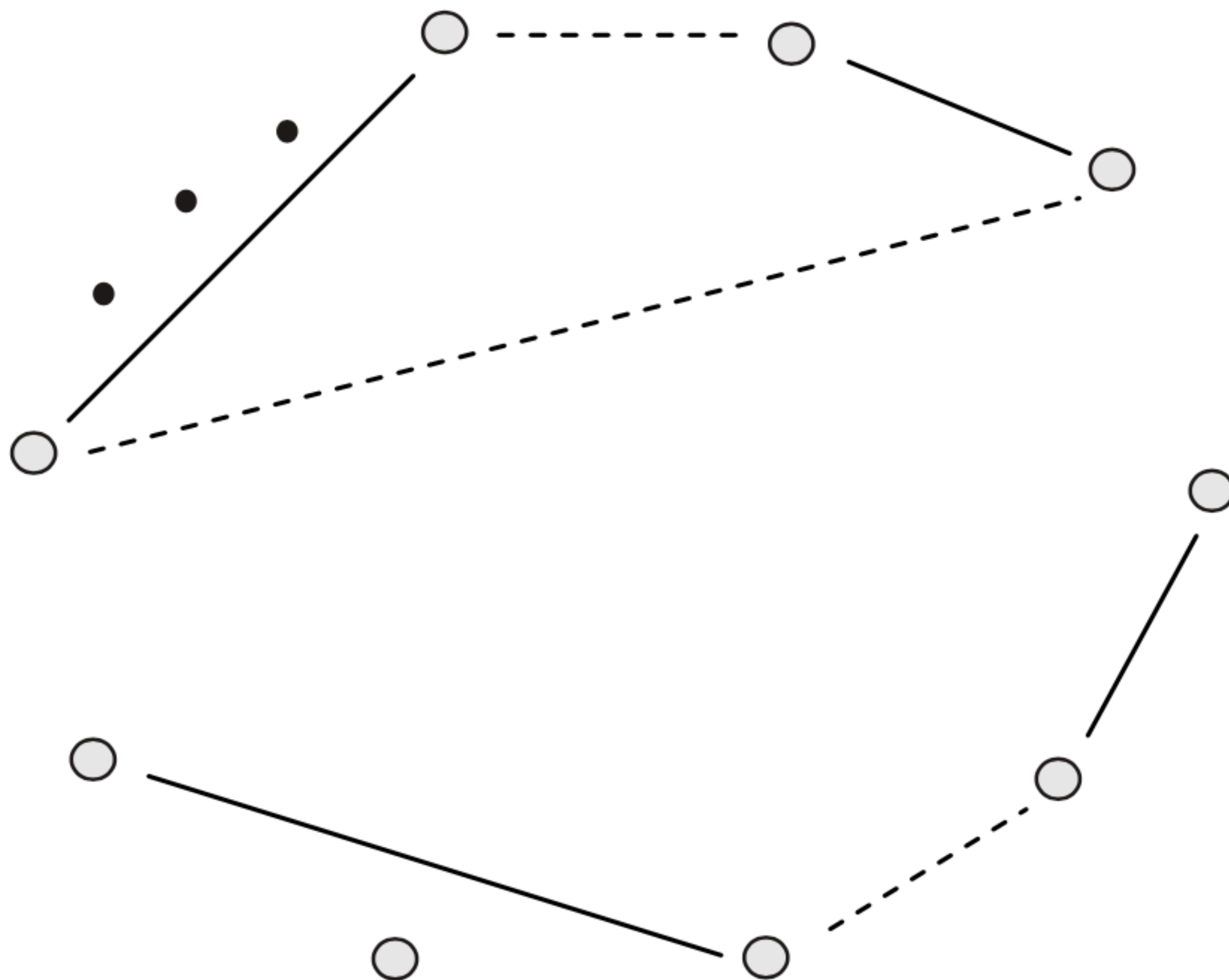


Circuit



Independent
Set

G

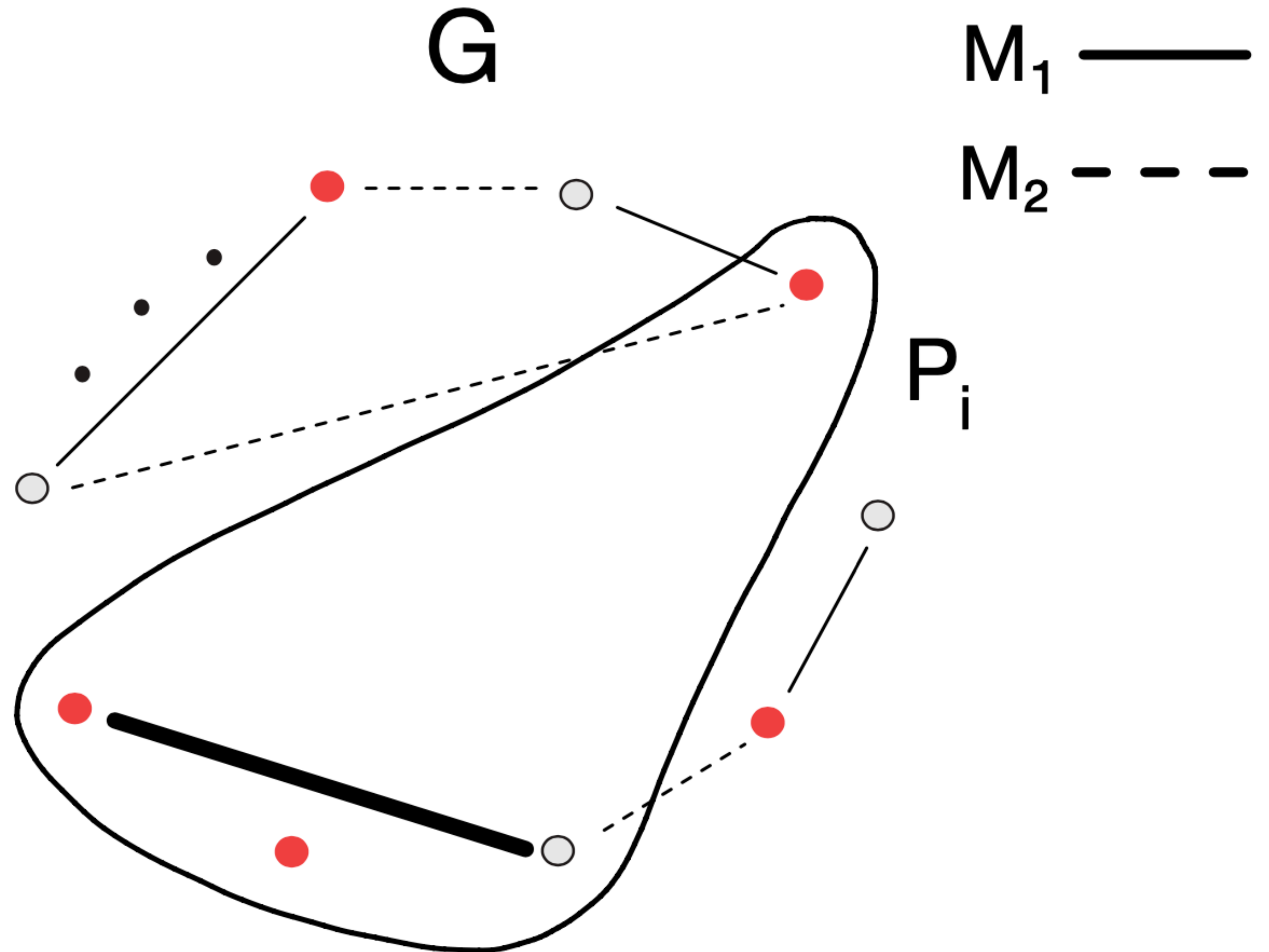


M₁ —

M₂ - - -

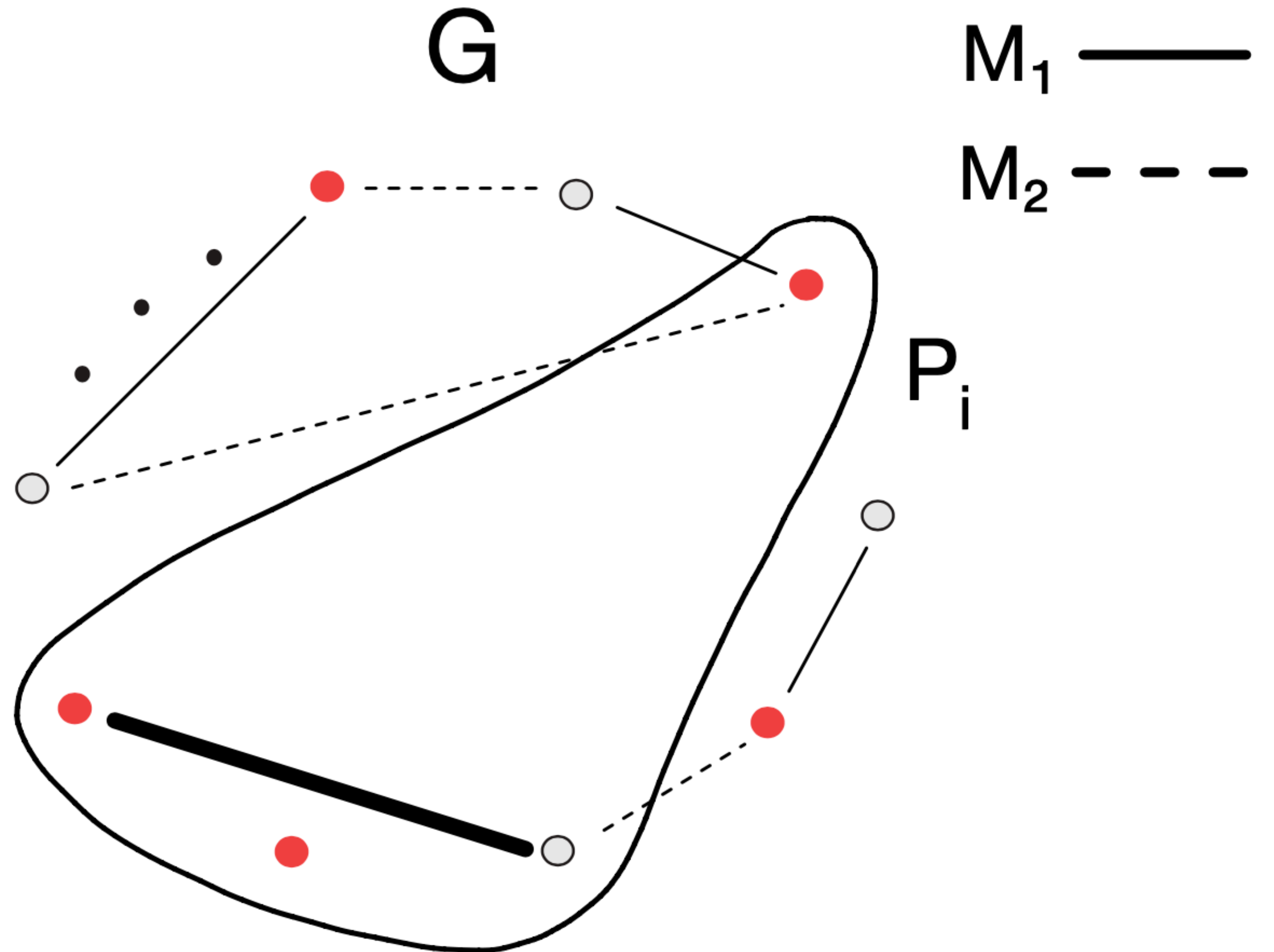
Lemma: A stable set in G is an independent set in $M_1 \cap M_2$.

- Let $T \subseteq S$ be a stable set in G .
- Consider part $P_i, i \in [k]$.
- **Claim:** $T \cap P_i$ is independent in M_1 .
- P_i independent ✓
- P_i circuit $\implies |T \cap P_i| < |P_i|$ ✓



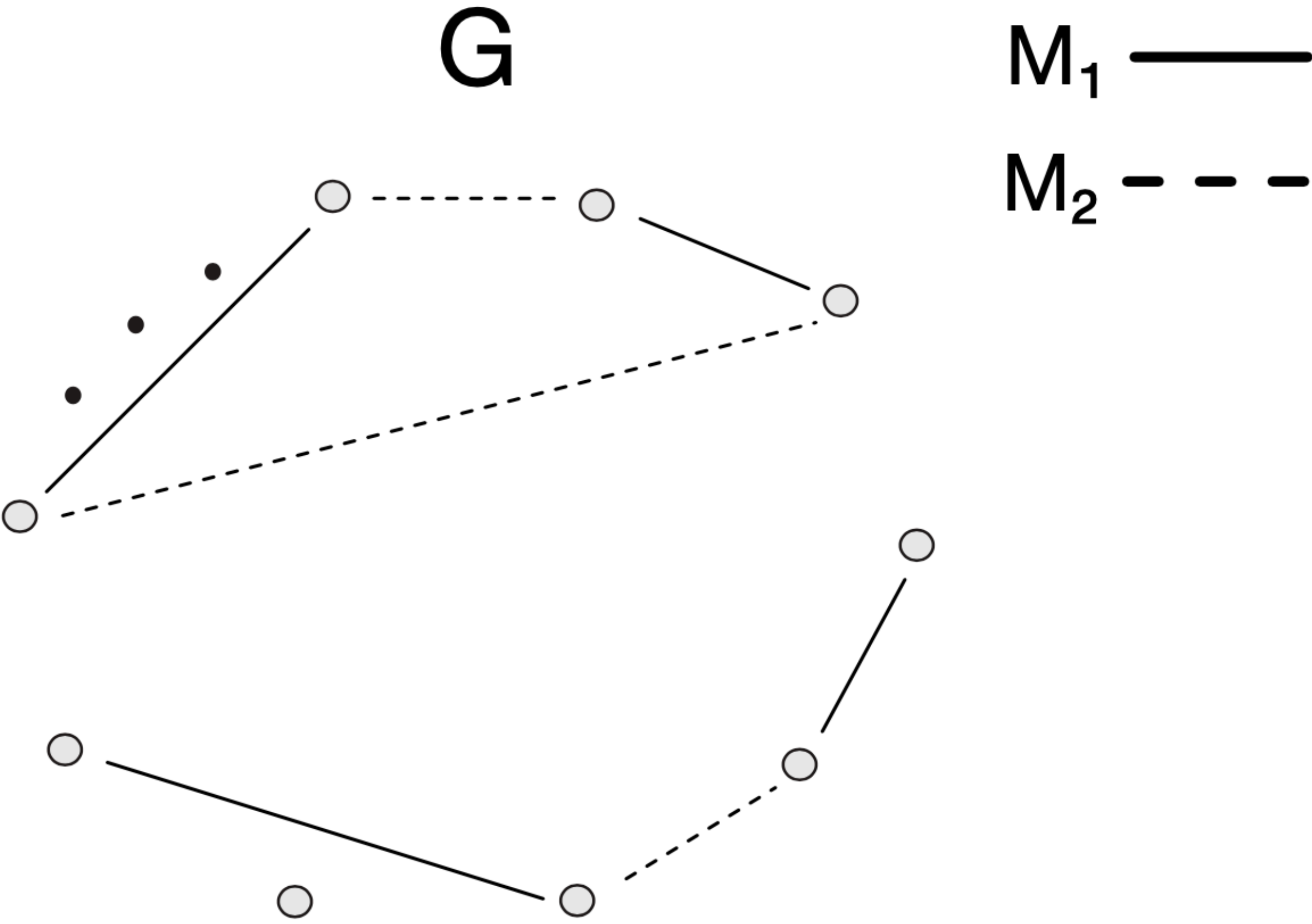
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- Let $T \subseteq S$ be a stable set in G .
- Consider part $P_i, i \in [k]$.
- **Claim:** $T \cap P_i$ is independent in M_1 .
- **Flexible Decomposition:**
 $\bigcup (T \cap P_i) = T$ is independent in M_1 . ✓



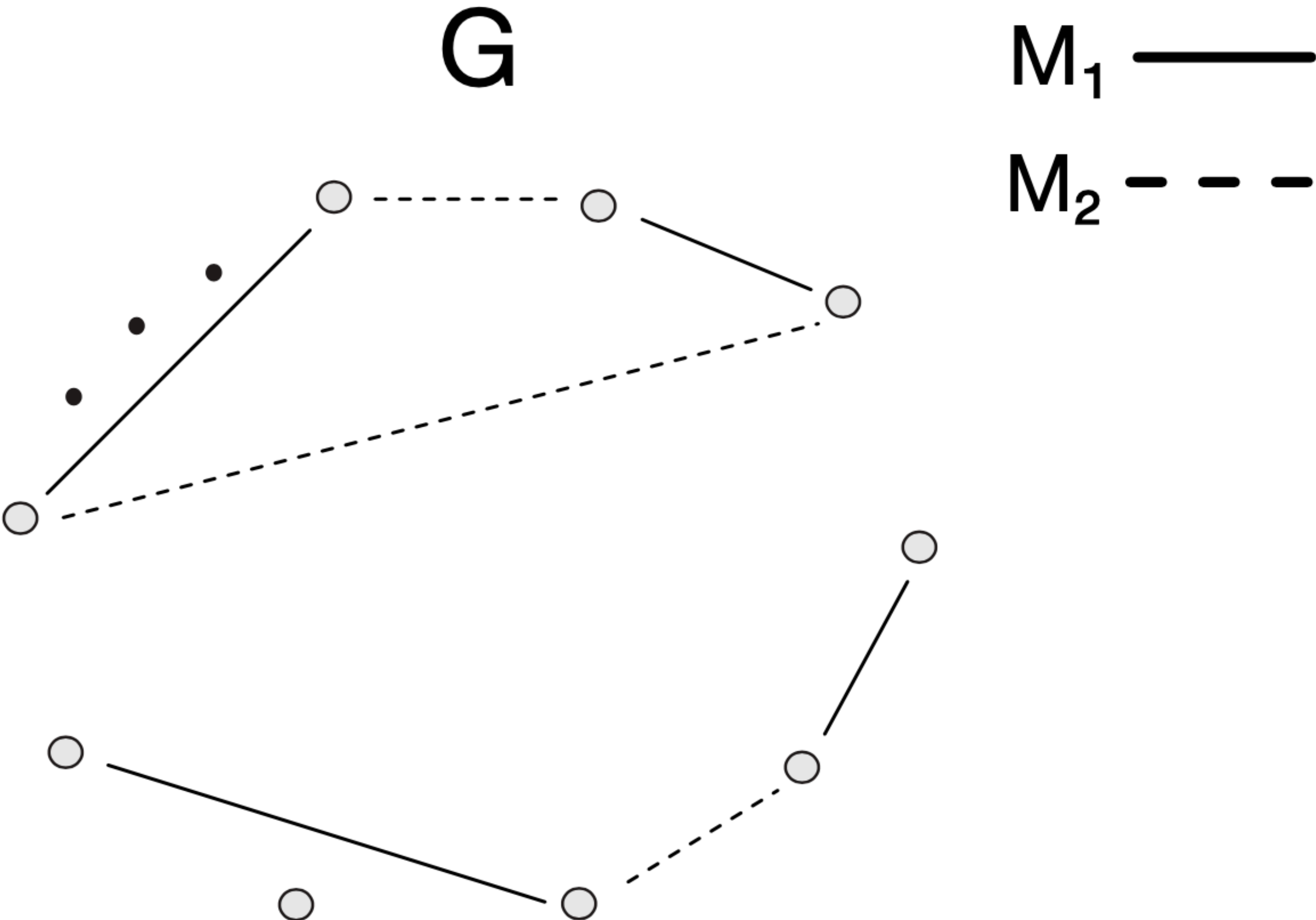
Lemma: G is 2-colorable.

Proof: G is the union of two matchings.



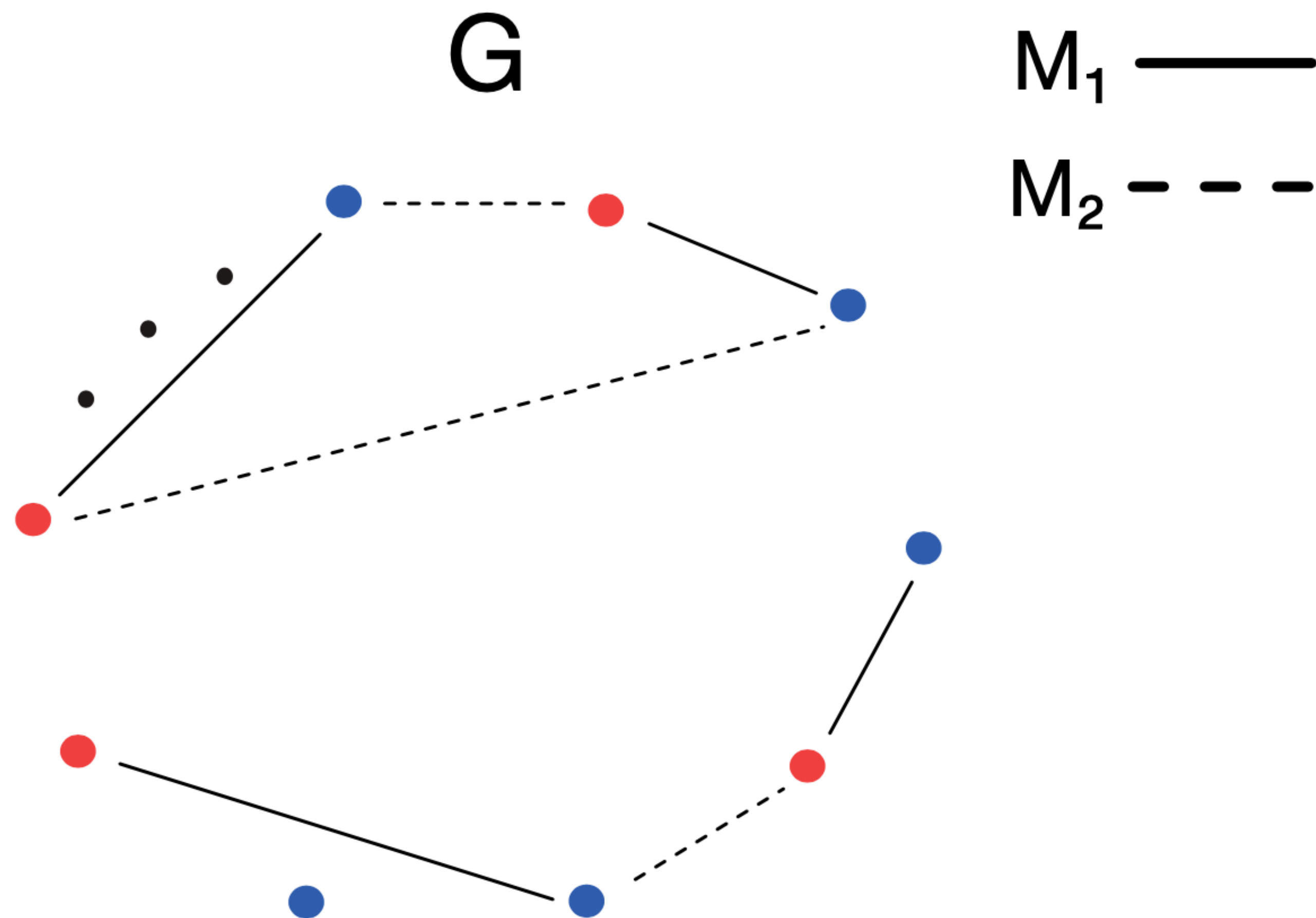
Lemma: G is 2-colorable.

Proof: Thus G decomposes into **disjoint paths** and **even cycles**.



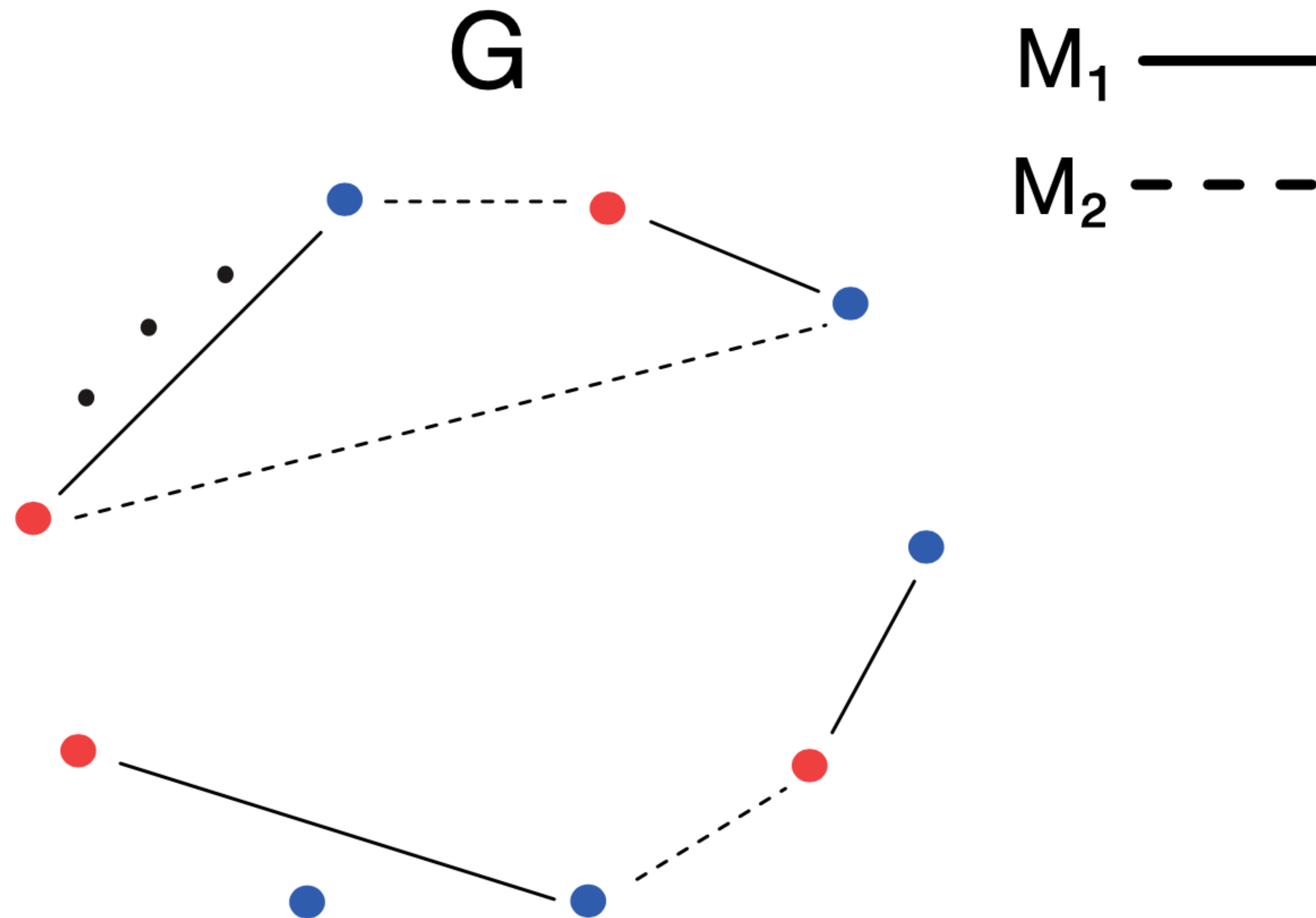
Lemma: G is 2-colorable.

Proof: Thus G is 2-colorable.



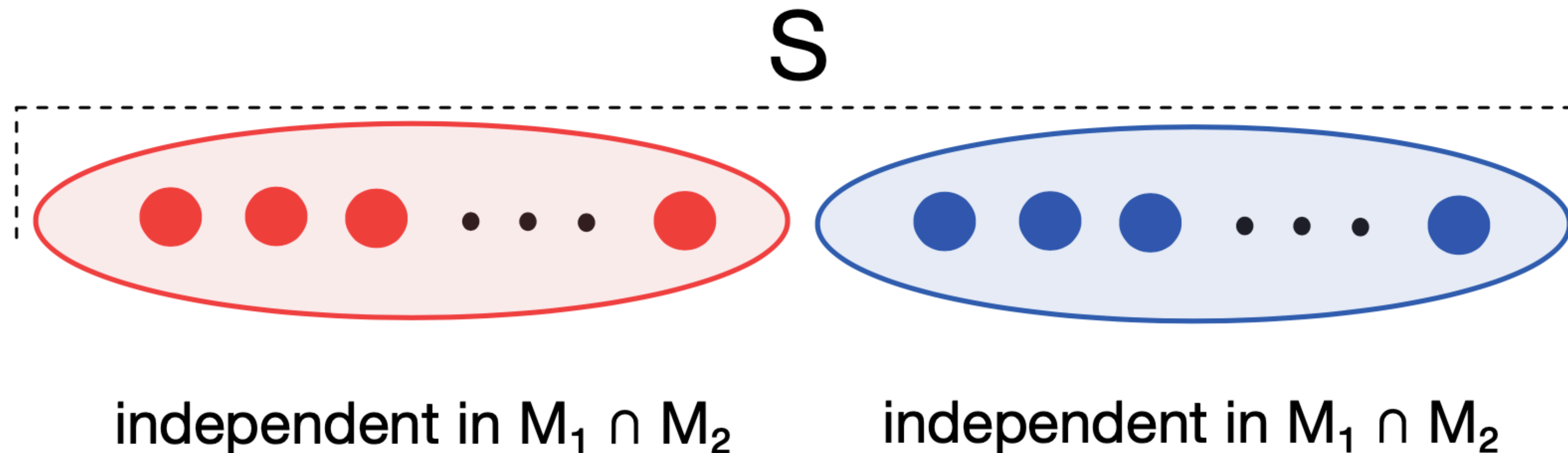
Proof of Main Theorem

- **Goal:** Replace each **pseudo color class** with 2 **valid color classes**.



Proof of Main Theorem

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Proof of Main Theorem

- **Goal:** Replace each **pseudo color class** with 2 **valid color classes**.

$\max\{\chi(M_1), \chi(M_2)\}$ -**pseudocoloring** of $M_1 \cap M_2$



$2 \max\{\chi(M_1), \chi(M_2)\}$ -**valid-coloring** of $M_1 \cap M_2$



- See paper for how to get **pseudocoloring** via **iterative rounding**!

Asymptotic MICP and Rota's Basis Conjecture

Rota's Basis Conjecture

Basis: Max-Size Independent Set
--

- **Def:** A set T is a **transversal** of $S_1, S_2, \dots, S_k$ if $|T \cap S_i| \leq 1$ for all $i \in [k]$.
- **Rota's Basis Conjecture ([HR94]):**
 - **Given:** General matroid M with r disjoint bases $B_1, B_2, \dots, B_r$ of size r .
 - **Claim:** There exist r disjoint bases $C_1, C_2, \dots, C_r$ of size r which are **transversals** of the original bases $B_1, B_2, \dots, B_r$.
- **Note:** We assume M has r^2 elements.

RBC as Matroid-Intersection Coloring

- **Conj ([HR94]):** For a **general** matroid M_1 with r disjoint bases $B_1, B_2, \dots, B_r$ of size r , and a **partition** matroid M_2 with parts $= B_i$, we have $\chi(M_1 \cap M_2) = r$.
- **Takeaway:** RBC is a special case of Matroid-Intersection Coloring!

Partition Matroid:
Independent Set means
 ≤ 1 per part

RBC as Matroid-Intersection Coloring

- **Conj ([HR94]):** For a **general** matroid M_1 with r disjoint bases $B_1, B_2, \dots, B_r$ of size r , and a **partition** matroid M_2 with parts $= B_i$, we have $\chi(M_1 \cap M_2) = r$.
- **Note:** $\chi(M_1) = r$ and $\chi(M_2) = r$.
- **Conj (High-Level):** For the RBC special case of Matroid-Intersection Coloring, where $\chi(M_1) = r$ and $\chi(M_2) = r$, we have $\chi(M_1 \cap M_2) = r$.

Understanding the Literature

- Two Main Research Lines
 1. Packing Many Bases of $M_1 \cap M_2$
 - 2. Covering by Few Independent Sets of $M_1 \cap M_2$**
 - Proofs about upper bounding $\chi(M_1 \cap M_2)$

Understanding the Literature

- **Thm** ([**AB06**]): $\chi(M_1 \cap M_2) \leq 2r$.
- **Thm** ([**Cho17**]): $\chi(M_1 \cap M_2) \leq 2r - 2$.
- **Thm** ([**MS25**]): $\chi(M_1 \cap M_2) \leq (1 + o(1))r$.
- **Research Question 2:** Does there exist a **constructive** proof (polynomial-time algorithm) for showing $\chi(M_1 \cap M_2) \leq (1 + o(1))r$ in the Rota's Basis Setting?

Rota's Basis
Special Case

Our Contribution

Our Contribution

- **Thm:** For two **general** matroids M_1 and M_2 such that

$$\max\{\chi(M_1), \chi(M_2)\} \geq \omega(\log n)$$

there is a **polynomial-time** algorithm that computes a

$$(1 + o(1)) \max\{\chi(M_1), \chi(M_2)\}\text{-coloring of } M_1 \cap M_2.$$

- **Application to RBC:** M_1 and M_2 have $r^2 = n$ elements and chromatic number $r = \sqrt{n}$.

Our Contribution

- **Thm:** For two **general** matroids M_1 and M_2 such that

$$\max\{\chi(M_1), \chi(M_2)\} \geq \omega(\log n)$$

there is a **polynomial-time** algorithm that computes a

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- Our result is **constructive**, handles **two general matroids**, gives **explicit asymptotics**, and is **simple**.

Our Contribution

- **Thm:** For two **general** matroids M_1 and M_2 such that

$$\max\{\chi(M_1), \chi(M_2)\} \geq \omega(\log n)$$

there is a **polynomial-time** algorithm that computes a

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- **Surprising:** Large chromatic number is the key!

Our Contribution

- **Thm:** For two **general** matroids M_1 and M_2 such that

$$\max\{\chi(M_1), \chi(M_2)\} \geq \omega(\log n)$$

there is a **polynomial-time** algorithm that computes a

$$(1 + o(1)) \max\{\chi(M_1), \chi(M_2)\}\text{-coloring of } M_1 \cap M_2.$$

- **Polyhedral** approach combined with **randomization**

More Background

Rank $r(S)$:
Size of largest
independent set in S

- **Thm ([Edm68]):** For a matroid $M = (E, \mathcal{S})$,

$$\chi(M) = \max_{S \subseteq E} \left\lceil \frac{|S|}{r(S)} \right\rceil$$

- $\chi(M)$ controlled by **maximum density** subset $S \subseteq E$.

A Natural Approach

1. Pick and remove some color classes.
 2. Argue that the maximum density went down in M_1 and M_2 .
 3. Repeat.
- **Goal:** m color classes drops max density by m .
 - **Issue:** Exponentially many subset densities to track! 
 - Reason why nonconstructive asymptotic result is complicated

A Simple and Flawed Approach

1. Sample and remove elements independently with probability p .
2. Maximum density drops by $\approx p \cdot \chi$ if $\chi \geq \omega(\log n)$.
3. Repeat.
 - Uniform hitting of every subset $S \subseteq X$ by **concentration!**
 - **Flaw:** Random sampling doesn't generate independent sets...
 - **Q:** Can we “simulate” random sampling?

Swap-Rounding Algorithm [CVZ11]

- **A:** Yes!
- **SwapRound:** Rounds **fractional points** $x \in P(M_1) \cap P(M_2)$ to **independent sets** $I \in \mathcal{S}_1 \cap \mathcal{S}_2$ s.t.
 1. **Marginal Values:** $\Pr[e \in I] \approx x_e$ for all elements $e \in E$.
 2. **Concentration:** Chernoff-style concentration bounds on the distribution of $|S \cap I|$ for all subsets $S \subseteq E$.
- **Takeaway:** SwapRound(x) simulates random sampling via x !

Our Simple and Correct Approach

Let $\chi_{\max} = \max\{\chi(M_1), \chi(M_2)\}$.

1. Sample and remove m sets via **SwapRound**($1/\chi_{\max}$) m times.
2. Maximum density drops by $\approx p \cdot \chi_{\max}$ if $\chi_{\max} \geq \omega(\log n)$.
3. Repeat.
 - Step 1 “simulates” **random sampling** using $p = m/\chi_{\max}$.
 - Thus maximum density drops by $\approx p \cdot \chi_{\max} = m$. ✓

Conclusion

- **Thm:** For two **general** matroids M_1 and M_2 ,

$$\chi(M_1 \cap M_2) \leq 2 \max\{\chi(M_1), \chi(M_2)\}$$

- If one matroid has **large chromatic number**,

$$\chi(M_1 \cap M_2) \leq (1 + o(1)) \max\{\chi(M_1), \chi(M_2)\}$$

- Further, both results are **constructive**.

Open Question: k General Matroids?

- **Thm** ([**ABGK25**]): For k general matroids $M_1, M_2, \dots, M_k$,

$$\chi\left(\bigcap M_i\right) \leq (2k - 1) \max_i \{\chi(M_i)\}$$

Linear in k

- **Thm**: For k general matroids $M_1, M_2, \dots, M_k$,

$$\chi\left(\bigcap M_i\right) \leq (k^2 - k) \max_i \{\chi(M_i)\}$$

Quadratic in k

Open Question: k General Matroids?

- **Thm** ([**ABGK25**]): For k general matroids $M_1, M_2, \dots, M_k$,

$$\chi\left(\bigcap M_i\right) \leq (2k - 1) \max_i \{\chi(M_i)\}$$

Linear in k

- **Q:** Does there exist a **constructive** proof of

$$\chi\left(\bigcap M_i\right) \leq O(k) \max_i \{\chi(M_i)\}$$

Linear in k



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Questions?